

VISHNU ROHIT ANNADANAM

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EDUCATION

Master of Science, Robotics | Northeastern University, Boston, MA | GPA – 3.8/4 Dec 2024

Bachelor of Engineering, Electronics & Telecommunication | University of Mumbai, India | GPA – 8.4/10 Jun 2022

EXPERIENCE

Robotics 3D Perception Engineer Intern | RIOS Intelligent Machines, Menlo Park, CA Jul 2025 – Present

- Calibrated Time-of-Flight (ToF) cameras on Fanuc-based autonomous industrial robots using custom acquisition scripts, achieving a 33% improvement in perception system precision.
- Developed automated ground-truth testing framework to collect and analyze the placement accuracy via perception-driven robot motions, eliminating manual measurements and accelerating validation workflows.
- Streamlined and optimized the ROS-based perception pipeline by adding real-time health monitoring and debugging utilities, reducing diagnosis time and improving overall system reliability.

Robotics Research Assistant [incl. Co-op] | Field Robotics Lab, Northeastern University, Boston, MA Sep 2023 – Jul 2025

- Led the design and development of a multi-UAV swarm system for mapping dynamic scenes, with a dual RGB camera setup for stereo on each drone and ensuring precise time synchronization across all cameras and platforms.
- Developed full-stack autonomy architecture in ROS2 integrating 3D object localization and multi-UAV formation control using *Python* & *C++* for tracking dynamic targets with collision avoidance and real-time camera feedback.
- Calibrated camera intrinsic and extrinsic parameters for the stereo-camera rig before each mission and used online calibration to get accurate depth estimates in real-time for optimal path planning, 3D reconstruction and mapping.
- Generated 3D/4D representations and novel views from multi-view datasets of dynamic scenes using *NeRFs* and *Gaussian Splatting*, from optimized sparse point clouds produced by Structure from Motion (SfM) algorithms.
- Benchmarked real-time Visual SLAM algorithms such as *NVIDIA ISAAC ROS VSLAM* on edge devices like Jetson Orin Nano, validating localization performance under compute and bandwidth constraints.

Graduate Teaching Assistant – Robotics Sensing and Navigation | Northeastern University, Boston, MA Sep 2023 – Dec 2024

- Mentored students on building device drivers, data post-processing and analysis for IMU, GNSS, RTK GPS, Camera and LiDAR sensors in ROS/ROS2, and guided them in deploying state-of-the-art SLAM algorithms on physical robots.

Computer Vision & IoT Intern | EdVerb Pvt. Ltd, Mumbai, India Jan 2022 – Jun 2022

- Programmed the *Arduino Uno* and *ESP-32* board and interfaced with multiple sensors to devise several cost-efficient computer vision and IoT-based solutions for automation.
- Designed and developed an automated parking management system using the *ESP-32 Cam* to implement automatic number plate detection and Optical Character Recognition (OCR) to register license plates.
- Integrated ultrasonic sensors for marking and determining slot availability at each parking space and continuously update the real-time database.

TECHNICAL SKILLS

Operating Systems & Software: ROS/ROS2 (Robot Operating System), Linux, CMake, Git, Docker, Gazebo, Raspberry Pi.

Programming Languages & Libraries: Python, C++, MATLAB, OpenCV, Open3D, PyTorch, CUDA, GTSAM, NumPy, Matplotlib.

Proficiencies: Robotics, Perception, SLAM, Sensor Fusion, Deep Learning, Motion Planning, System Design, CAD, 3D Printing.

PROJECTS

GPS & IMU Sensor Fusion using Extended Kalman Filter (EKF)

- Built ROS2 drivers to collect GNSS & IMU sensors data, analyzed IMU noise via Allan Variance, calibrated magnetometer distortions & accelerometer bias, and fused magnetometer & gyroscope heading estimates for robust dead reckoning.
- Performed GPS and IMU sensor fusion by implementing Extended Kalman Filtering (EKF) to get an improved estimate of vehicle's overall trajectory including GPS-denied environments.

Semantic Segmentation of Road Cracks using Swin Transformer ([link](#))

- Trained a surface crack segmentation model for autonomous road inspection using Swin Transformer + UPerNet architecture, utilizing *PyTorch* and *MMSegmentation*, with ADE20K pre-trained weights for improved transfer learning.
- Achieved ~78% mIoU on the test set at 80k iterations by employing a hybrid loss strategy combining Cross-Entropy loss and Lovasz-Softmax loss to optimize IoU on fine-grained crack patterns across a dataset of 11.2K images.

3D Point Cloud Reconstruction from 2D Images through Structure from Motion (SfM) ([link](#))

- Developed a 3D reconstruction pipeline using Structure from Motion, generating sparse point clouds from 2D images with *OpenCV* for feature extraction (*SIFT* + *RANSAC*) and initial triangulation, and *GTSAM* for optimization.
- Estimated camera poses and triangulated 3D points using essential matrix and PnP pose computation, and performed global Bundle Adjustment using *GTSAM* factor graphs to refine poses & landmarks, reducing reprojection error by 94.5%.

PUBLICATIONS

- A System for Multi-View Mapping of Dynamic Scenes Using Time-Synchronized UAVs, *IROS 2025*.